

Math 104 – Real Analysis

Homework 3 — my solutions

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Solutions

Problem 1

Solution. We start by dividing off by n^6 to hopefully manipulate this nicely:

$$\frac{11n^6 - 4n^5 + 9n^2 - 7}{13n^6 + 5n^4 + 2n + 1} = \frac{11 - 4\frac{1}{n} + 9\frac{1}{n^4} - 7\frac{1}{n^6}}{13 + 5\frac{1}{n^2} + 2\frac{1}{n^5} + \frac{1}{n^6}}$$

Let's examine the numerator:

Notice, $\lim_{n \rightarrow \infty} (11 - 4\frac{1}{n} + 9\frac{1}{n^4} - 7\frac{1}{n^6}) = 11$ because

$$\lim_{n \rightarrow \infty} \left(11 - 4\frac{1}{n} + 9\frac{1}{n^4} - 7\frac{1}{n^6} \right) = \lim 11 - 4 \lim \frac{1}{n} + 9 \lim \frac{1}{n^4} - 7 \lim \frac{1}{n^6} = 11$$

using $\lim_{n \rightarrow \infty} \frac{1}{n^p} = 0$ for $p > 0$. This exact same argument can be applied for the denominator where everything but the 13 goes to zero because:

$$\lim_{n \rightarrow \infty} \left(13 + 5\frac{1}{n^2} + 2\frac{1}{n^5} + \frac{1}{n^6} \right) = \lim 13 + 5 \lim \frac{1}{n^2} + 2 \lim \frac{1}{n^5} + \lim \frac{1}{n^6} = 13 + 5(0) + 2(0) + 0 = 13$$

And the quotient rule applies because the denominator limit is nonzero. Thus, this limit evaluates to $\frac{11}{13}$. $\square$

Problem 2

Solution. We are given the following:

$$\lim_{n \rightarrow \infty} x_n = -2, \lim_{n \rightarrow \infty} y_n = 5$$

and that $x_n + 2y_n \neq 0$ for every $n \in \mathbb{N}$.

Because the denominator will always be nonzero as given, we can consider separately the limit of the numerator and denominator, and apply the quotient rule because the limit of the denominator is also nonzero.

When considering

$$\lim_{n \rightarrow \infty} (2y_n - x_n) = 2 \lim y_n - \lim x_n = 2(5) - (-2) = 12.$$

To consider the denominator, notice

$$\lim_{n \rightarrow \infty} (x_n + 2y_n)^2 = \left(\lim_{n \rightarrow \infty} (x_n + 2y_n) \right)^2$$

from the product rule, and so we have that this limit is equal to

$$(\lim x_n + 2 \lim y_n)^2 = (-2 + 10)^2 = 64$$

So our answer is $\frac{12}{64} = \frac{3}{16}$. □

Problem 3

Solution. We first notice that

$$\lim_{n \rightarrow \infty} (a_n^2 + b_n^2 + 1) = a^2 + b^2 + 1 \geq 1 > 0,$$

so the limit of the denominator is nonzero.

So in order to continue, let's consider the numerator and denominator separately. We have

$$\lim_{n \rightarrow \infty} (a_n^4 - 2a_n b_n + 3) = \lim a_n^4 - 2 \lim a_n \lim b_n + \lim 3 = a^4 - 2ab + 3$$

Considering the denominator similarly, notice that

$$\lim_{n \rightarrow \infty} (a_n^2 + b_n^2 + 1) = (\lim a_n)^2 + (\lim b_n)^2 + 1 = a^2 + b^2 + 1$$

so we are done, as we have shown that

$$\lim_{n \rightarrow \infty} u_n = \frac{a^4 - 2ab + 3}{a^2 + b^2 + 1}.$$

□

Problem 4

Solution. We have that $x_1 = 1$ and $x_{n+1} = x_n(4x_n - 1)$. Notice we can write: We claim $x_n \geq 3^{n-1}$ for all positive integers n , and will show this by induction.

The base case is trivial as $x_1 = 1 \geq 3^{1-1} = 1$.

For the inductive step, assume that $x_k \geq 3^{k-1}$ for some positive integer k . Then,

$$x_{k+1} = x_k(4x_k - 1).$$

Let's look at our inductive hypothesis:

$$x_k \geq 3^{k-1} \geq 1 \rightarrow 4x_k \geq 4 \rightarrow 4x_k - 1 \geq 3.$$

So, we have that $4x_k - 1 \geq 3$ and we are done because then

$$x_{k+1} \geq 3^{k-1} \cdot 3 = 3^k.$$

So by induction this is true for all $k \in \mathbb{Z}^+$. We have that $x_n \geq 3^{n-1}$ and $\lim_{n \rightarrow \infty} 3^{n-1} = +\infty$, and so by comparison, $\lim_{n \rightarrow \infty} x_n = +\infty$. $\square$

Problem 5

Solution. We have that $r_n \rightarrow 0$ and $\exists N_0 \in \mathbb{N}$ such that

$$s_n \leq t_n + r_n \text{ for every } n > N_0.$$

We write $t_n \geq s_n - r_n$ and just wish to show rigorously that " s_n " gets large and r_n is very small.

Consider an arbitrary $M > 0$. Since $s_n \rightarrow +\infty$, there exists $N_2 \in \mathbb{N}$ such that for every $n > N_2$,

$$s_n > M + 1.$$

We write

$$t_n \geq s_n - r_n.$$

Along with this, we notice since $r_n \rightarrow 0$, there exists $N_1 \in \mathbb{N}$ such that $|r_n| < 1$ for every $n > N_1$, so notice that for every $n > \max\{N_0, N_1, N_2\}$,

$$t_n \geq s_n - r_n > s_n - 1 > M + 1 - 1 = M.$$

So for arbitrary M there exists $N = \max\{N_0, N_1, N_2\}$ such that for every $n > N$, both $s_n > M + 1$ and $|r_n| < 1$ hold, which satisfies $t_n > M$. Therefore,

$$\lim_{n \rightarrow \infty} t_n = +\infty.$$

$\square$

Problem 6

Solution.

$\square$

Problem 7

Solution.

$\square$

Problem 8*Solution.*

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